Executive Briefing & Facility Vulnerability Assessment

FORENSIC IDENTIFICATION STATEMENT: 100% UNREGISTERED / UNKNOWN SUBJECTS
Zero facial profiles match an enrolled employee whitelist or authorized staff database. **All 1,652 detected personas (SUBJ-01 to SUBJ-1652) are formally classified as UNKNOWN PERSONS (Pending Identification).** Sighting frequency tiers denote statistical recurrence volume only and carry no implication of employment authorization or access clearance.

1. Executive Mission Overview

An autonomous forensic and biometric surveillance audit was conducted over 30 continuous calendar days (713.66 recorded hours) of native 3K CCTV footage covering the RO Water Treatment Plant rooftop mechanical deck (Camera IPC56). The audit cataloged every recurring human identity, traced operational shift patterns, and documented all physical contacts against high-hazard equipment assets.

2. Verified Operational & Security Metrics

Metric Category	Audit Result	Security & Operational Context
Continuous Facility Coverage	**713.66 Recorded Hours**	30 continuous calendar days (99.12% facility uptime across 999 clips)
Total Discovered Personas	**1,652 Unique Personas**	100% Unknown / Unregistered Individuals grouped via ArcFace clustering
High-Frequency Unknowns (Tier 1)	**12 Individuals**	Recurring unknown subjects with 50+ detections each
Periodic Unknowns (Tier 2)	**85 Individuals**	Recurring unknown subjects with 10 - 49 detections
Transient Unknowns (Tier 3)	**1,555 Individuals**	Low-frequency unknown subjects with 2 - 9 detections
Direct Equipment Contact Breaches	**18 Unknown Individuals**	Verified touching multiport valves, skid controls, or electrical breaker switch
Lossless Incident Clips	**179 Events (8.20 GB)**	Lossless H.265 stream cuts with +/-60s contextual pre/post buffers
Processed Face Vectors	**36,226 Embeddings**	512-D ArcFace vectors passing strict quality threshold (Q >= 40)

3. Operational Shift Workload Distribution

Shift A (Morning, 06:00 – 13:59): 24,625 detections (68.0%) — Primary operational window where recurring unknown subjects routinely operated valves and inspected the RO skid.
Shift B (Afternoon, 14:00 – 21:59): 11,188 detections (30.9%) — Secondary window with multiple unknown subjects traversing the terrace corridor.
Shift C (Night Vigilance, 22:00 – 05:59): 413 detections (1.1%) — Off-hours access events including perimeter walkthroughs and main breaker switch contacts.

4. 30-Day Critical & Severe Incident Investigations (Days 01 – 30)

Across the 30 continuous calendar days, **179 physical equipment interaction events** were verified and cataloged across three hazard tiers:

Day & Timestamp	Event ID	Facility Asset	Unknown Subjects Involved	Duration
Day 09 16:21:55	EVT-D09-005	Ultimate Power Switch (Zone 2)	**SUBJ-233;SUBJ-759;SUBJ-760**	162s
Day 09 16:24:51	EVT-D09-006	Ultimate Power Switch (Zone 2)	**SUBJ-13;SUBJ-347**	125s
Day 11 06:33:37	EVT-D11-001	Ultimate Power Switch (Zone 2)	**SUBJ-791**	121s
Day 24 16:39:26	EVT-D24-013	Ultimate Power Switch (Zone 2)	**SUBJ-1079**	121s
Day 25 11:43:10	EVT-D25-002	Ultimate Power Switch (Zone 2)	**SUBJ-1088**	121s

Critical Multiport Backwash Valve Operations (Zone 1A/B — 54 Events across 23 Days):
• **Week 1 (Days 01–07, 9 events):** Routine morning backwashes, including Day 1 events (SUBJ-01 at 07:29/08:14 and SUBJ-1652 at 09:53:38).
• **Week 2 (Days 08–14, 13 events):** Daily operational cycling by Tier 1 and Tier 2 unknown subjects.
• **Week 3 (Days 15–21, 17 events):** Peak critical activity window, featuring Day 15 multi-subject group access (5 valve operations involving 10 unknown subjects).
• **Week 4 (Days 22–26, 15 events):** Frequent high-contact days including Day 25 (4 valve operations) and Day 26 mass gathering (3 events, 34 unknown subjects on deck).
• **Night Vigilance (Shift C):** Off-hours visits on Day 01 (22:27:36), Day 15 (23:24:04), and Day 29 (23:14:10).

5. Evidence Continuity & Forensic Quality Assurance

1. Rolling Continuous Temporal Windows: Replaced static segment cutoffs with rolling temporal windowing, eliminating blindspots across continuous 45-minute CCTV segments.
2. Contextual Safety Buffers (±60s): All 179 extracted incidents feature mandatory 60-second pre- and post-event safety margins that bridge video file splits without loss of visual context.
3. Biometric Separation & Enrollment (SUBJ-1652): Verified the Day 1 09:53:38 valve operator is distinct from SUBJ-02 (ArcFace cosine similarity 0.1831, far below the 0.4500 threshold). Formally enrolled as SUBJ-1652.
4. Chronological ISO 8601 Wall-Clock Ordering: Rebuilt all 30 clipping indices with strict chronological sorting, achieving 0 timestamp inversions across all 179 events.

6. Facility Security & Compliance Recommendations

1. Mandatory Personnel Enrollment: Execute a formal identification drive with facility administration to associate legal identities with all 1,652 unknown persona clusters (SUBJ-01 to SUBJ-1652).
2. Physical Access Restriction: Install Lockout/Tagout (LOTO) padlock enclosures on Zone 1A/B Multiport Valves and the Zone 2 Main Power Breaker Switch to prevent unrecorded manual operation.
3. Automated Spatial Gating Alarms: Deploy the calibrated spatial ROI mesh (IoU > 0.08) for real-time audible alarms whenever an unregistered individual approaches within 40px of critical equipment.